Taxonomy of Symbolic Structure and Autonomous Feedback

Distribution of the 35 included studies across two independent dimensions

Symbolic structure	Without autonomous feedback	With autonomous feedback
No symbolic structure	**Group A** Direct Generation 14 included studies (40.0%)	No included studies
Lightweight or static symbolic structure	**Group B** Static Neuro-Symbolic 16 included studies (45.7%)	No included studies
Formal symbolic structure	**Group C** Formal Verification 2 included studies (5.7%)	**Group D** Execution-Aware Recovery 3 included studies (8.6%)

Research gap: No included study demonstrates a complete closed-loop neuro-symbolic recovery cycle combining dynamic-world change, semantic diagnosis of physical failure, formal model updating, and autonomous replanning without possible human fallback.